

AMARILLO NWS, TX, USA

WMO: 723630

Lat: 35.2331N

Lon: 101.7092W

Elev: 1095

StdP: 88.84

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23047

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-8.9	-18.2	0.9	-3.2	-16.2	1.0	0.4	14.7	8.8	13.3	9.8	5.0	0	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	37.1	18.7	35.6	18.9	34.2	18.9	21.9	30.4	21.3	30.0	20.7	29.5	6.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	16.4	24.2	18.9	15.7	23.7	18.3	15.1	23.3	69.3	30.3	66.8	29.9	64.7	29.6	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.6	12.0	10.9	DB	-15.7	39.4	2.5	1.8	-17.5	40.7	-19.0	41.8	-20.4	42.8	-22.3	44.2	(4)
(5)			WB	-16.3	22.9	2.4	0.8	-18.0	23.5	-19.4	23.9	-20.7	24.4	-22.5	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	3.5	5.1	9.5	13.8	19.0	24.2	26.1	25.3	21.4	14.9	8.6	3.6		
	DBStd	9.44	5.40	6.10	5.65	5.04	4.63	3.40	2.64	2.80	3.92	4.88	5.33	5.42		
	HDD10.0	771	208	155	77	22	2	0	0	0	1	14	86	206		
	HDD18.3	2196	459	371	275	147	48	2	0	1	17	126	292	457		
	CDD10.0	2466	7	18	62	137	281	426	499	475	342	167	44	7		
	CDD18.3	848	0	0	2	12	69	178	241	217	108	19	1	0		
	CDH23.3	10171	1	14	96	294	965	2094	2887	2365	1137	282	36	1		
	CDH26.7	5003	0	2	21	89	437	1084	1558	1226	498	86	3	0		
(14) Wind	WSAvg	5.8	5.4	5.8	6.2	6.7	6.1	6.2	5.5	5.1	5.4	5.6	5.7	5.5		
(15) Precipitation	PrecAvg	508	16	15	28	31	60	86	72	78	49	39	18	15		
	PrecMax	931	80	64	105	164	236	273	204	231	126	165	103	63		
	PrecMin	182	0	0	0	0	1	3	1	7	1	0	0	0		
	PrecStd	134	17	14	27	33	44	59	49	51	33	35	19	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	25.3	29.4	31.8	36.4	39.6	38.8	38.1	36.1	31.7	26.9	21.8		
		MCWB	9.2	10.5	12.0	13.0	15.9	17.4	19.2	19.2	18.4	15.9	11.6	8.9		
	2%	DB	19.2	22.1	26.0	29.4	34.0	36.9	36.9	36.0	33.7	29.0	24.0	18.7		
		MCWB	7.8	9.2	11.3	12.5	15.6	17.6	19.5	19.4	18.1	15.3	11.4	7.9		
	5%	DB	16.0	19.1	23.3	26.8	31.5	34.7	35.4	34.5	31.8	26.8	21.0	15.9		
		MCWB	6.5	8.2	10.5	12.5	15.6	18.6	19.6	19.6	18.3	14.8	10.1	6.9		
	10%	DB	13.0	15.7	20.3	24.3	28.9	32.7	33.8	33.0	29.6	23.9	18.1	12.9		
		MCWB	5.3	6.6	9.3	12.0	15.4	18.7	19.7	19.5	17.9	13.8	9.1	5.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	11.9	15.2	17.4	20.3	22.4	22.9	22.5	21.3	19.5	14.6	11.4		
		MCDB	19.5	20.1	20.6	23.2	27.4	30.3	30.7	30.8	29.3	23.9	19.6	17.0		
	2%	WB	8.4	10.2	13.2	15.6	19.1	21.4	21.9	21.7	20.4	18.0	12.9	9.0		
		MCDB	17.3	19.8	20.8	22.2	26.8	29.8	30.5	30.4	28.3	23.2	18.9	16.9		
	5%	WB	6.9	8.8	11.7	14.4	18.2	20.7	21.2	21.1	19.7	16.9	11.5	7.4		
		MCDB	15.1	18.1	20.6	21.8	26.0	29.3	30.4	29.8	27.5	22.2	18.5	15.2		
	10%	WB	5.4	7.2	10.3	13.0	17.2	20.1	20.6	20.5	19.0	15.5	10.1	5.7		
		MCDB	12.6	15.3	18.9	21.6	25.0	28.6	30.1	29.1	26.5	21.2	17.5	12.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	14.9	15.6	15.5	14.7	14.3	13.8	13.4	13.6	14.2	14.5	13.6		
		MCDBR	19.3	20.2	19.9	19.5	18.6	17.2	15.9	15.4	16.3	17.8	18.8	18.4		
		MCWBR	10.4	10.3	9.0	8.2	7.1	5.4	4.4	4.4	5.4	7.3	9.0	10.0		
		MCDBR	17.9	18.5	17.6	15.9	15.3	14.0	13.6	13.2	13.3	13.1	15.9	17.3		
	5% WB	MCWBR	9.9	10.0	8.7	8.2	7.0	5.3	4.7	4.5	5.0	6.8	8.4	9.9		
(40) Clear-Sky Solar Irradiance	taub		0.253	0.261	0.290	0.326	0.338	0.378	0.381	0.370	0.340	0.299	0.271	0.256		
	taud		2.603	2.589	2.511	2.409	2.438	2.372	2.403	2.400	2.458	2.564	2.570	2.590		
	Ebn at Noon		975	999	987	957	944	902	897	903	918	939	941	949		
	Edh at Noon		73	84	99	116	114	122	118	116	104	85	75	70		
(44) All-Sky Solar Radiation	RadAvg		3.14	3.95	5.20	6.40	7.09	7.53	7.27	6.51	5.56	4.25	3.32	2.71		
	RadStd		0.27	0.35	0.37	0.39	0.54	0.45	0.31	0.32	0.29	0.46	0.33	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+112	N/A
(47) Regional (20 neighbors)		+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+96	N/A

Nomenclature: See separate page